

14/8/26

CS 340

lecture 7

Epsilon Transitions in FA

Automata with ϵ -transitions

So far, all transitions of our automata involve reading a letter of alphabet Σ . Epsilon transitions ~~but~~ in automata do not require reading a letter from Σ , they can happen ~~in~~ in the absence of any input, ^{there are} therefore labeled by ' ϵ ' (the empty string).

NFA transition fn $\delta: Q \times \Sigma \rightarrow 2^Q$

NFA with ϵ transitions $\delta: Q \times (\Sigma \cup \{\epsilon\}) \rightarrow 2^Q$

We need to extend the definition of run/computation to NFA with ϵ transitions.

Defn ($\xrightarrow{*}_\epsilon$)

Let δ be the transition fn of an ~~and~~ NFA with ϵ -transitions. We define $\xrightarrow{*}_\epsilon$ inductively as follows.

- (i) $q \xrightarrow{*}_\epsilon q \quad \forall q \in Q$
- (ii) $q \xrightarrow{*}_\epsilon q' \quad \text{if } q' \in \delta(q, \epsilon) \quad [\text{can be derived from (i) \& (ii)}]$
- (iii) if $q \xrightarrow{*}_\epsilon p$ and $p' \in \delta(p, \epsilon)$
then $q \xrightarrow{*}_\epsilon p'$

An alternative way of defining $\xrightarrow{*}_\epsilon$ is that $q \xrightarrow{*}_\epsilon q'$ iff there is a sequence of zero or more ϵ transitions as follows

$$q \xrightarrow{\epsilon} q_1 \xrightarrow{\epsilon} q_2 \xrightarrow{\epsilon} \dots \xrightarrow{\epsilon} q_k \xrightarrow{\epsilon} q'$$

(Hence $q \xrightarrow{\epsilon} p \iff p' \in \delta(p, \epsilon)$)

Yet another way of defining $\xrightarrow{*}$ is to take reflexive transitive closure of relation $p \xrightarrow{\epsilon} p'$.

Def (run/computation)

Let $M = (Q, \Sigma, q_0, \delta, F)$ be a NFA with ~~ϵ -trans~~ ϵ -transitions. A run of M on input $x = a_1 a_2 \dots a_k$ is defined as a sequence $q_0 q'_0 q_1 q'_1 \dots q_k q'_k$ of states ~~s.t.~~ s.t. $\forall i: 0 \leq i \leq k [q_i \xrightarrow{*} q'_i]$ and $\forall i: 1 \leq i \leq k [q'_{i-1} \xrightarrow{a_i} q_i]$.

More pictorially, a run of M on $x = a_1 a_2 \dots a_k$ can be shown as

$$q_0 \xrightarrow{*} q'_0 \xrightarrow{a_1} q_1 \xrightarrow{*} q'_1 \xrightarrow{a_2} q_2 \rightarrow \dots \xrightarrow{a_k} q_k \xrightarrow{*} q'_k$$

Basically, it is like a run of NFA but now from any state ϵ -transitions may be made for free.

A run is accepting, if the last state of the run $\in F$.

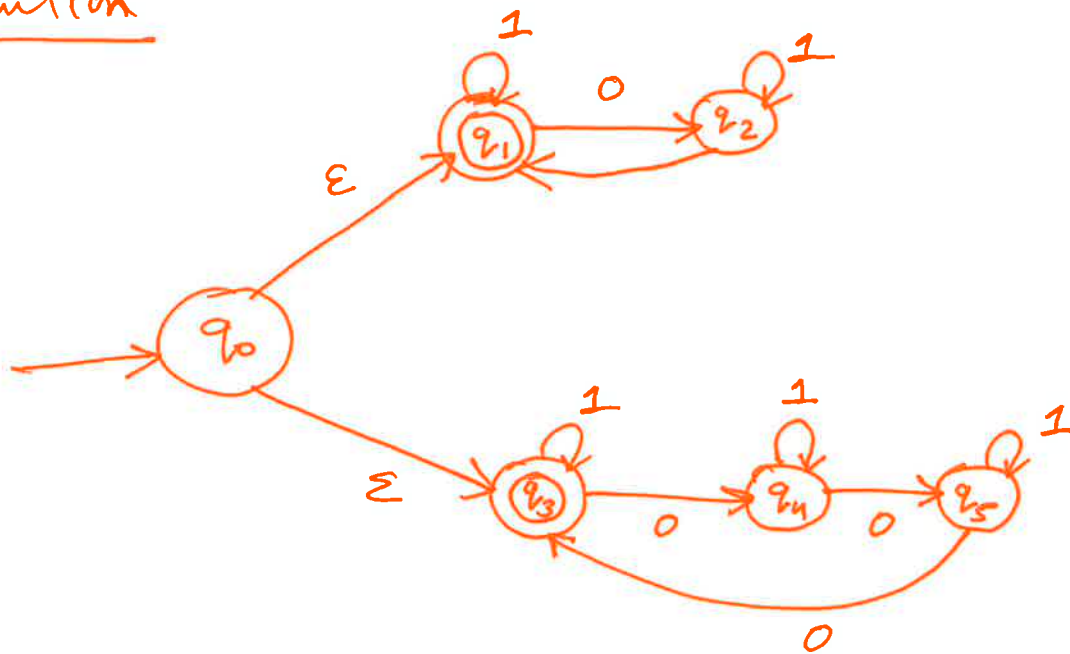
$$L(M) = \{x \in \Sigma^* \mid \text{M has an accepting run on } x\}$$

Let us give some examples of NFAs with ϵ transitions.

Example 1:

Design an automaton (NFA with ϵ transitions, if needed) which accepts $L = \{x \in \{0,1\}^* \mid \begin{matrix} \text{no. of 0's in } x \\ \text{are either } 0 \bmod 2 \text{ or } 0 \bmod 3. \end{matrix} \}$

Solution

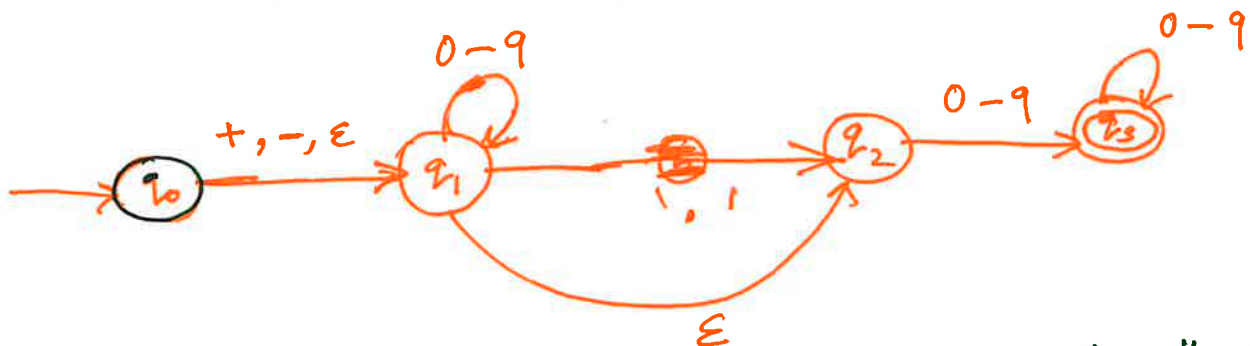


While automaton for language L above may be designed using product automata construction for the union operation, above automaton is simpler to understand.

Example 2:

Consider description of real no. literals in some programming language:

- (i) Option $+/-$ sign in the beginning
- (ii) At least one digit in the literal
- (iii) An optional decimal point followed by at least one digit.



Above NFA ~~recognizes~~ (with ϵ transitions) recognizes the real literals of a ~~hypothetical~~ hypothetical programming language.

The next theorem shows a general construction ~~for~~ of equivalent DFA for any ~~given~~ NFA with ϵ transitions. ~~The~~ Following definition is helpful in the construction.

Def (ϵ -closure)

Let $M = (Q, \Sigma, q_0, \delta, F)$ be a NFA with ϵ -transitions. For any $q \in Q$, we define

$$\epsilon\text{-closure}(q) = \{q' \in Q \mid q \xrightarrow{\epsilon}^* q'\}$$

In state diagram representation of M , $\epsilon\text{-closure}(q)$ is simply the set of states reachable from q using ϵ -transitions alone.

Theorem-1:

For every NFA with ϵ -transitions there is a DFA which accepts the same language. Further, if no. of states in NFA (with ϵ -transitions) is n then the DFA has at most 2^n states.

Proof:

Let $M = (Q, \Sigma, q_0, \delta, F)$ be a NFA with ϵ -transitions. We define equivalent DFA D as

$$D = (2^Q, \Sigma, \epsilon\text{-closure}(q_0), \delta_D, F')$$

where $\delta_D(\{p_1, \dots, p_k\}, a) = \bigcup_{i=1}^k \epsilon\text{-closure}(\delta(p_i, a))$

(Here ϵ -closure is extended to sets in a natural way as $\epsilon\text{-closure}(S) = \bigcup_{p \in S} \epsilon\text{-closure}(p)$, for $S \subseteq Q$)

$$\delta_D(\{p_1, \dots, p_k\}, a) = \bigcup_{i=1}^k \bigcup_{q \in \delta(p_i, a)} \varepsilon\text{-closure}(q)$$

and $F = \{F' \subseteq Q \mid F' \cap F \neq \emptyset\}$

Let $q \xrightarrow[x]{M}^* q'$ stand for "there is a computation of M on input x starting in state q and ending in state q' "

(Note that computation of M may use ε -transitions at any step.)

We now claim that

$$\delta_D(q_{0D}, x) = \{q' \in Q \mid q_0 \xrightarrow[x]{M}^* q'\}$$

($q_{0D} = \varepsilon\text{-closure}(q_0)$ = initial state of D)

proof of the claim is by induction on $|x|$.

Base Case: $x = \varepsilon$

$$\begin{aligned} \delta_D(q_{0D}, \varepsilon) &= q_{0D} \\ &= \varepsilon\text{-closure}(q_0) \\ &= \{q' \mid q_0 \xrightarrow[\varepsilon]{M}^* q'\} \end{aligned}$$

(by def of $\hat{\delta}_D$)

(by defn of q_{0D})

(by defn of $\varepsilon\text{-closure}$)

*/

Induction Step:

$$x = wa$$

$$\begin{aligned}
 \delta_D(q_0, wa) &= \delta_D(\delta_D(q_0, w), a) && \text{(by def of } \hat{\delta}_D) \\
 &= \delta_D(\{q' \mid q_0 \xrightarrow{w}_M^* q'\}, a) && \text{(By I.H.)} \\
 &= \bigcup_{q' : q_0 \xrightarrow{w}_M^* q'} \bigcup_{p \in \delta(q', a)} \varepsilon\text{-closure}(p) && \text{(By def of } \delta_D) \\
 &= \bigcup_{q' : \dots} \bigcup_{p \in \delta(q', a)} \{r \in Q \mid p \xrightarrow{\varepsilon}_M^* r\} && \text{(By def of } \varepsilon\text{-closure)} \\
 &= \{r \in Q \mid \exists q' [q_0 \xrightarrow{w}_M^* q' \wedge \exists p \in \delta(q', a) (p \xrightarrow{\varepsilon}_M^* r)]\} && \text{(By def of 'U')} \\
 &= \{r \in Q \mid \exists q' \exists p [q_0 \xrightarrow{w}_M^* q' \xrightarrow{a}_M p \xrightarrow{\varepsilon}_M^* r]\} \\
 &= \{r \in Q \mid q_0 \xrightarrow{wa}_M^* r\} && \text{(By def of } \xrightarrow{y}_M^*)
 \end{aligned}$$

This complete induction step and proof of the claim.

Now,

$$\begin{aligned}
 x \in \cancel{L(D)} &\Leftrightarrow \delta_D(q_0, x) \in F \\
 &\Leftrightarrow \delta_D(q_0, x) \cap F \neq \emptyset \\
 &\Leftrightarrow \exists q \in Q [q_0 \xrightarrow{x}_M^* q \wedge q \in F] && \text{(By claim above)} \\
 &\Leftrightarrow \text{there is an accepting computation of } M \text{ on input } x \\
 &\Leftrightarrow x \in L(M)
 \end{aligned}$$

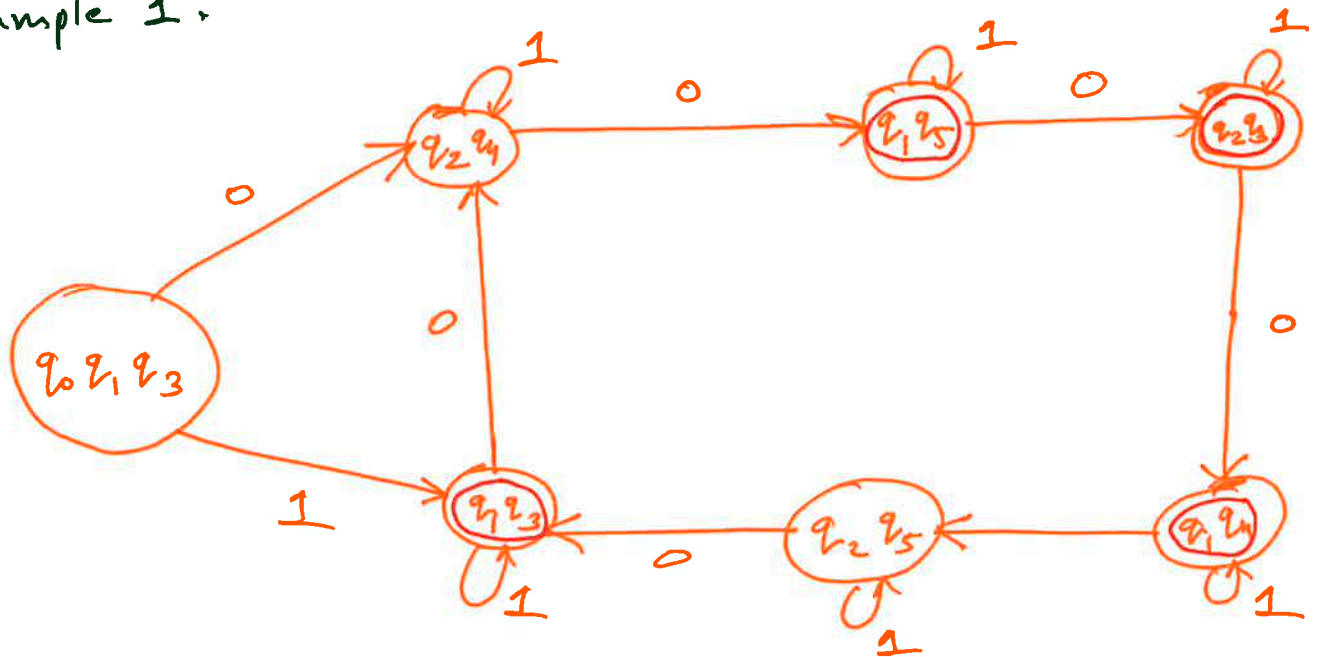
□

Let us run the construction in Theorem 1 on examples of ϵ -automata considered earlier.

Example 3:

We construct DFA for the automaton in

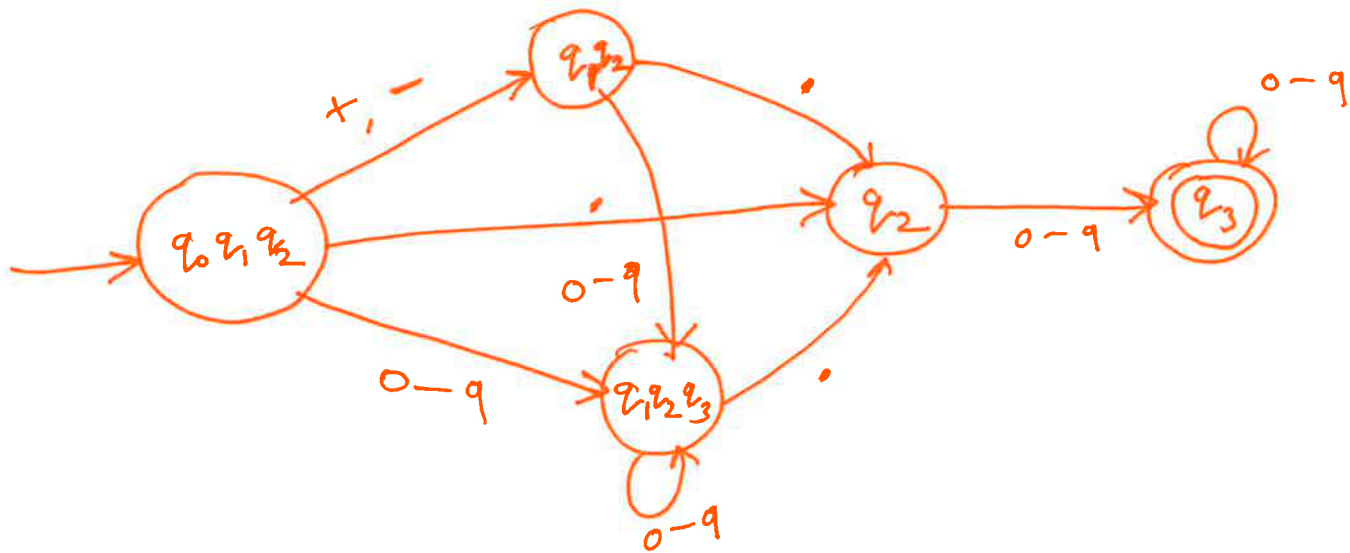
Example 1.



Note that after the initial state above DFA is the same as the product DFA constructed in the Union construction.

Example 4

We construct DFA for the automaton in Example 2.



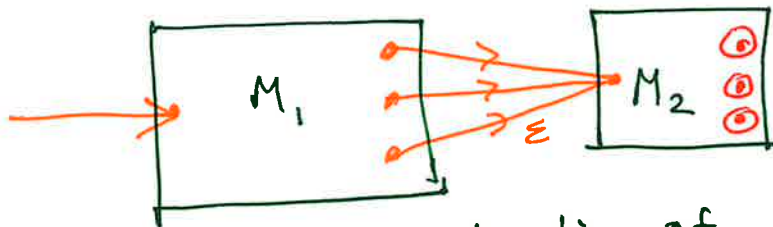
We have shown that DFA, NFA, ~~NFA~~ NFA with ϵ transitions recognize the same class of languages. This allows us to show that this class is closed under ~~Concatenation~~ Concatenation and Kleene closure operations as follows.

Example 5

closure under ' . '

Recall that $L_1 \cdot L_2 = \{x_1 x_2 \mid x_1 \in L_1, x_2 \in L_2\}$

Initial state of M is the initial state of M_1



Construction of M

Final states of M are the final states of M_2

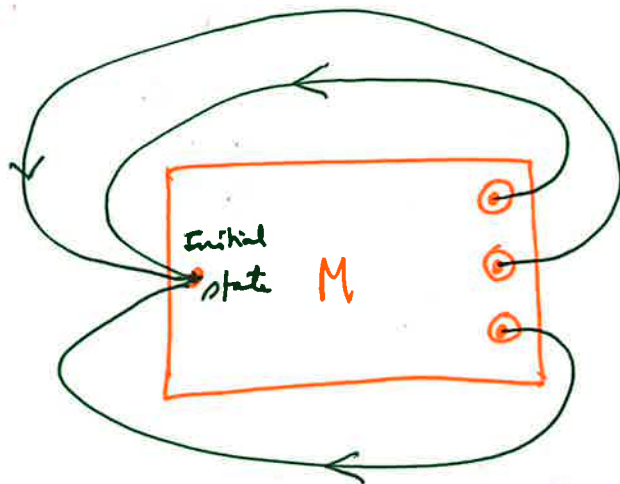
M_1, M_2 are NFAs with ϵ transitions
 $L(M) = L(M_1) \cdot L(M_2) \quad \square$

Example 6

Closure under $*$ (Kleene star)

Recall that $L^* = \{x_1 \dots x_n \mid \text{for } n \geq 0, \text{ s.t. } x_1, x_2, \dots, x_n \in L\}$

Initial state
and final states
of M, M'
are the same.



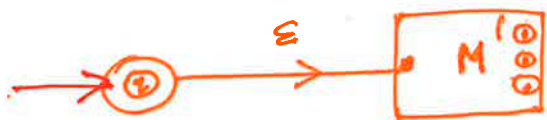
$$L(M') = L^+$$

Construction of M'

For L^* , we also need to have ϵ in the language

$$L^* = L^+ \cup \{\epsilon\}$$

Exercise: Show that using a direct transition from initial state of M to a final state of M may not result in the correct construction.



Automaton to recognize
 L^*

Construction of M'' .

Final states are q (new state, not in M')
and the final states of M' .